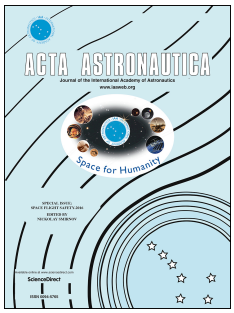
Accepted Manuscript



Experimental research on the spray characteristics of pintle injector

Huiyuan Chen, Qinglian Li, Peng Cheng

PII: S0094-5765(19)30607-1

DOI: https://doi.org/10.1016/j.actaastro.2019.06.032

Reference: AA 7576

To appear in: *Acta Astronautica*

Received Date: 1 April 2019

Revised Date: 15 May 2019

Accepted Date: 25 June 2019

Please cite this article as: H. Chen, Q. Li, P. Cheng, Experimental research on the spray characteristics of pintle injector, *Acta Astronautica* (2019), doi: https://doi.org/10.1016/j.actaastro.2019.06.032.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Experimental research on the spray characteristics of pintle injector

Huiyuan Chen[1,2], Qinglian Li[1,2*], Peng Cheng[1,2]

(1. Science and Technology on Scramjet Laboratory, National University of Defense Technology, Changsha 410073, China.

2. College of Aerospace and Engineering, National University of Defense Technology, Changsha 410073, China)

Abstract

The spray characteristics of an oxidizer-centered liquid-liquid pintle injector at different momentum ratios are experimentally investigated. Water is used as the simulation fluid of both the fuel and the oxidizer and injected into the axial annular gap and radial orifices, respectively. The radial-to-axial local momentum ratio (LMR) is varied from 0.16 to 0.93 by adjusting the mass flow rate of the oxidizer under constant mass flow rate of the fuel. A high-speed camera and particle dynamics analysis (PDA) system are used to investigate the spray pattern, spray angle, droplet size and velocity distribution. The results indicate that the spray generated by the pintle injector is shaped as a hollow-to-solid cone with a rough boundary. A hollow structure appears in the spray center in the near field and disappears in the far field. With the increase of the momentum ratio, the outer boundary of the spray field is enlarged and the range of the hollow field also increases. The spray angle varies from 34° to 122° for the LMR in the range of 0.16 ~ 0.93. Furthermore, the spray angle is deduced as a monotonically increasing function of the local momentum ratio of the pintle injector with discrete radial orifices. The diameter-velocity distribution indicates that SMD (Sauter

* Corresponding author: peakdreamer@163.com
E-mail address: catherinechy@163.com

Mean Diameter) increases slightly and the axial velocity decreases mildly with the increase in the momentum ratio. The SMD curve exhibits an "*N*" shape in the radial direction, whereas the axial velocity and radial velocity curves exhibit inverted "*V*" shapes in general. The spray distribution and spray development are mainly controlled by the interaction of the radial flow of the jet and the axial flow of the film, and also to some degree by the impingement of the liquid droplets in the spray center. From the spray center to the edge, the spray of the pintle injector is divided into three regions: the spray core, the mainstream, and the outer boundary.

Keywords: pintle injector; spray pattern; spray angle; diameter-velocity distribution; spray development

1. Introduction

The pintle injector is developed and used exclusively in liquid rocket engine applications [1]. The spray from a pintle injector is formed by the collision of radial jets of fluid originating from the center of the pintle post with an annular gap of fluid traveling axially along the post. The radial jets can be ejected from discrete orifices or from a continuous gap.

The pintle injector concept was pioneered by the TRW Corporation, which is now represented by the Northrop Grumman Space Technology (NGST) Division of the Grumman Corporation. Reference [2] sums up the development of the pintle engine in the early days. The lunar module descent engine (LMDE) is one of the most well-known applications of the pintle injector technology. In the engine, a movable sleeve was used to control both the annular and orifice areas to provide the 10:1 throttling ratio (from 10000 to 1000 lbf of thrust) required by this application [3, 4]. The pintle injector is also applied in the investigations of nontoxic hypergolic

propellants [5-8].

Most of the previous studies on pintle injectors have focused on the spray angle. Son et al. [9] conducted a research on the effects of the gas-to-liquid momentum ratio and Weber number on the spray characteristics under various injection conditions for an injector with a radial gap. The spray half angles had an exponentially decreasing correlation with the total momentum ratio (TMR) divided by the Weber number. A subsequent simulation [10] also showed that the spray angle decreased with the increase in the gas-to-liquid momentum ratio. Cheng et al. [11] conducted a cold test and numerical simulation to investigate the spray angle resolution under different operating conditions and put forward a prediction equation of the spray half angle: $\cos \alpha=1/(1+TMR)$. The prediction results agreed well with the experimental results. Santoro and Merkle et al. [12, 13] investigated the cold spray of a pintle injector with discrete radial orifices and determined that the spray angle increased with the increase in the radial to axial momentum ratio. In addition, the influence of the blockage factor (BF) was not apparent for an injector with a BF between 0.3 and 0.8. Nevertheless, Heister [1] considered the BF a key parameter.

In spite of the great importance, the droplet size and mixing characteristics of the pintle injector have not been investigated comprehensively. Son et al. [14] measured the SMD for different Weber numbers and radial opening distances, discovering that the SMD exhibited the same trend as the radial opening distance and the opposite trend as the Weber number. Son et al. [10] also measured the spray density along the radial direction by collecting the spray and conducting a numerical simulation. The experimental and simulation results were in agreement. Radhakrishnan et al. [15] researched on the influence of the injection conditions on the mixing performance. A parameter $N_k=1/(1+TMR)$ was used to represent the injection condition. It was concluded that the dispersion

angle and the mixing quality decreased and the spray angle increased with the increase in N_k. The breakup and evaporation process is also of great importance, but there has been no work directly focused on that of the pintle injector, while the investigations [16-18] of Nickolay N. et al. are enlightening.

To sum up, many studies on the spray characteristics of the pintle injector have been conducted, but these studies are not comprehensive. The spray angle has been covered relatively well and a consensus has been reached that the spray angle increases with the increase in the radial to axial momentum ratio. A prediction formula has been deduced for the pintle injector with a radial gap. However, only limited investigation has been conducted on the distribution of the droplet size and the mixing performance. Therefore, more systematic research is still required. In this study, a liquid-liquid pintle injector is designed. The spray pattern, spray angle, and diameter-velocity distribution are investigated by conducting a cold test under different momentum ratios.

2. Experimental equipment and data processing methods

The experimental setup is depicted in Fig. 1. Apart from the basic composition like supply system, measurement and control system and the pintle injector, the image acquisition system and particle dynamics analysis (PDA) system are also equipped to record the spray development and measure the particle size and velocity.

An oxidizer-centered liquid-liquid pintle injector with a mass flow rate of 0.7 kg/s (493 g/s for the fuel and 206 g/s for the oxidizer) was designed and investigated. The designed oxidizer and fuel are LOX and ethanol individually, while water is used as the simulant fluid for both of the oxidizer and the fuel in this paper. The fuel travels downstream before colliding with the oxidizer and the distance is defined as the “skip distance”[1]. It is noted that the inlet of the outer pintle is tangential.

ACCEPTED MANUSCRIPT

The schematic diagram of the pintle injector is shown in Fig. 2 and the detailed parameters of the pintle injector are listed in Table 1. The blockage factor is an important design variable, defined as the ratio of total hole/slot circumferential length divided by the circumference of the pintle

$$BF = \frac{nd_t}{\pi D} \tag{1}$$

where n and d_t are the number and the diameter of the orifices in the pintle tip, respectively. D is the diameter of the pintle tip.

ACCEPTED MANUSCRIPT

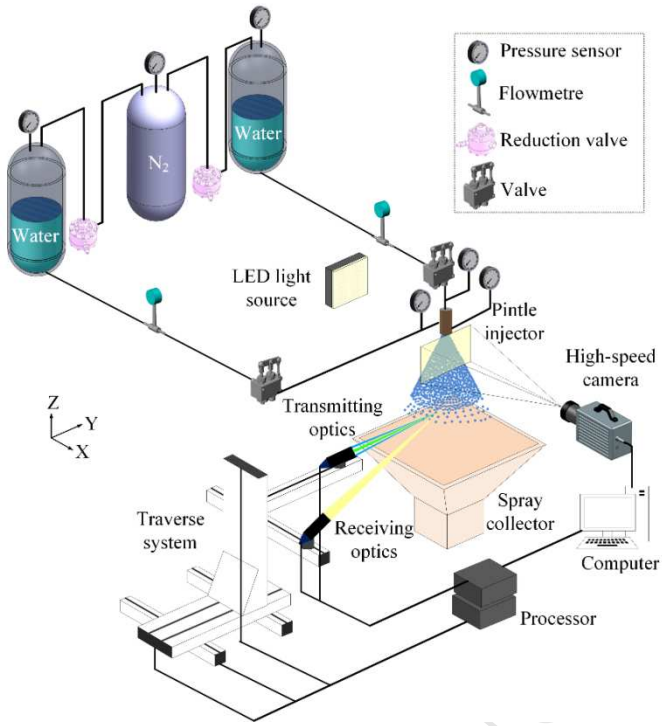


Fig. 1: Schematic of the experimental setup [19]

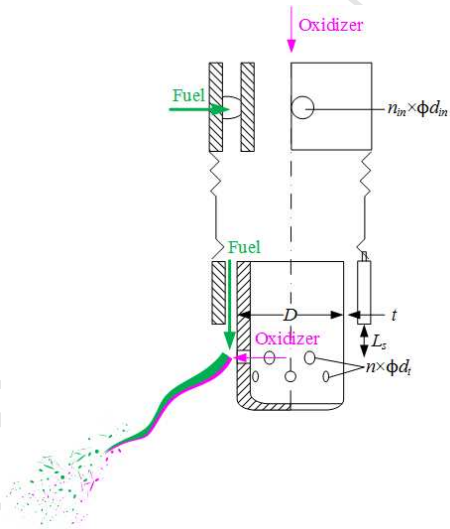


Fig. 2: Schematic diagram of the pintle injector

Table 1: Structural parameters of the pintle injector

Pintle tip diameter D/mm	10.4
Orifice rows n_r	2
Orifice diameter d_t/mm	1.1/0.9
Orifice number n	6/6
Skip distance L_s/mm	10.4
Annular gap thickness t/mm	0.5
Blockage factor (BF)	0.3673

Number of the tangential orifices n_{in}	4
Diameter of the tangential orifices d_{in}/mm	2.9

2.1 Experimental conditions

The spray characteristics for different momentum ratios are investigated by varying the mass flow rate of the oxidizer under a constant mass flow rate of the fuel. Considering that the jet is ejected as multi-strand jets, the local field generated by the interaction of single jet and the responding film is concerned. Table 2 displays the operating conditions, including the mass flow rate of the inner/outer flow, the total momentum ratio (TMR) and the local momentum ratio (LMR).

Table 2: Operating conditions

Operating conditions	Outer mass flow rate/(g/s)	Inner mass flow rate/(g/s)	Total momentum ratio (TMR)	Local momentum ratio (LMR)
gk1	502	59.3	0.1116	0.1550
gk2	503	120	0.1892	0.3662
gk3	505	149	0.2436	0.5143
gk4	505	209	0.3956	0.9283

Owing that the outer flow is ejected through the tangential orifices, the liquid film carries both of the axial and radial momentum, which should be considered while calculating the radial-to-axial momentum ratio. In order to calculate the component momentum of the film in the axial and radial direction, the film is injected alone in advance and the spray angle is measured and written as α_o. Furthermore, the component momentum ratio of the film in the axial and radial direction are $\dot{m}_o v_o \cos(\alpha_o/2)$ and $\dot{m}_o v_o \sin(\alpha_o/2)$, respectively. Thus, the TMR is defined as the ratio of the radial-to-axial stream momentum

$$TMR = \frac{\dot{m}_2 v_2}{\dot{m}_1 v_1} = \frac{\dot{m}_i v_i + \dot{m}_o v_o \sin(\alpha_o/2)}{\dot{m}_o v_o \cos(\alpha_o/2)} \tag{2}$$

where $\dot{m}$ and v represent the mass flow rate and velocity respectively. The subscript "1" and "2" indicate the axial and radial direction, and the subscript "*i*" and "*o*" indicate the inner flow and

outer flow, respectively.

And the local momentum ratio is given in Table 2. It is defined as

$$LMR = \frac{\dot{m}_{2l} v_2}{\dot{m}_{1l} v_1} = \frac{\dot{m}_i v_i + BF \cdot \dot{m}_o v_o \sin(\alpha_o / 2)}{BF \cdot \dot{m}_o v_o \cos(\alpha_o / 2)} \tag{3}$$

where the subscript "*l*" represents the flow parameter of the interaction between a single radial orifice and the responding film.

2.2 Image acquisition system

The image acquisition system is used to visualize the spray field, which includes a high-speed camera, an LED light source, and a computer. The LED light source provides a stable background light and is used to visualize the flow field. The images are recorded by a high-speed camera (Photron FASTCAM SA-X2) equipped with a Nikon ED 200 mm prime lens. The frame frequency and exposure time are 10000 fps and 1/100000 second, respectively. The image resolution is 784×958 pixels.

2.3 PDA system

The optical PDA system consists of the transmitting optics and the receiving detector, which are mounted on a three-dimensional coordinate location device. The deviation angle of the transmitting optics and the receiving detector is 145°. A fiber is used to connect the processor and computer; the optical signal is processed, providing information on the droplet size, velocity, and other data. The green laser beam is used to measure the droplet size and axial velocity, while the blue laser beam is used to measure the x-direction velocity, which is also the radial velocity at the XOZ plane. It is noted that the configuration is set as 2D PDA non-coincidence to obtain sufficient data at either direction, but the statistical results can be used to characterize the spatial distribution

of the spray characteristics. For each measurement point, the maximal samples and maximal acquisition time are set as 10000 and 10.0 seconds, respectively. The PDA system will move to the next measurement point when either condition is satisfied. The parameters of the PDA system are displayed in Table 3. More detailed information about the PDA system can be found in Wu [20].

The spray field under different operating conditions is visualized by the image acquisition system firstly and the spray range is estimated to decide the PDA measurement grid, as shown in Fig. 3. Considering the construction of the pintle injector is quasi-axisymmetric, the measurement points of the operating condition gk2 are set in a cylindrical coordinate system to obtain the three-dimensional distribution of the spray field, while a longitudinal section is set as the characteristic plane under other operating conditions. There are 46/52/65/79 measurement points at the characteristic plane under the operating conditions gk1-gk4, respectively. The origin is located at the center of the pintle tip bottom. The coordinate transformation between the cylindrical coordinate and Cartesian coordinate is

$$\begin{cases} x = r \times \cos\theta \\ y = -r \times \sin\theta \\ z = h \end{cases} \tag{4}$$

Table 3: Parameters of the PDA system

Laser wavelength (nm)	514.5/488
Off-axis angle (deg.)	145
Axial velocity range (m/s)	-7~45
Radial velocity range (m/s)	-28~46
Diameter range (μm)	0.1-1000
Focal length of the transmitting optics (mm)	500
Focal length of the receiving detector (mm)	1000

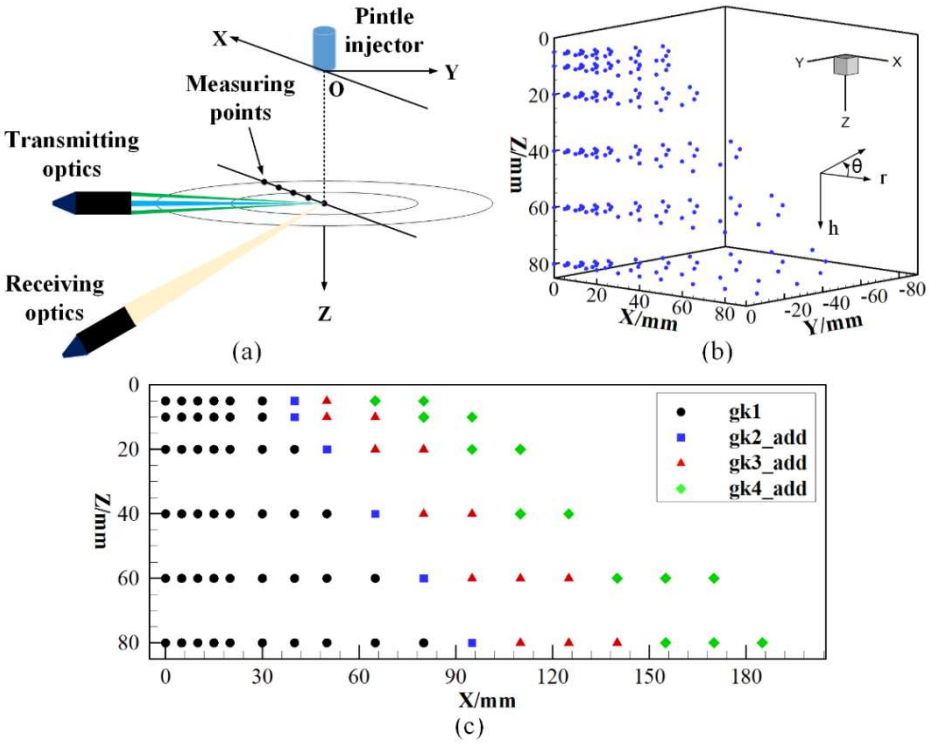


Fig. 3: PDA system (a) and measurement point setup; (b) gk2 (3-D); (c) gk1-gk4 (2-D)

2.4 Data processing method

As the background light passes through the spray, the light intensity is attenuated by the scattering and absorption of the spray, which results in an inhomogeneous gray level distribution in the images: the denser the spray, the lower the gray level. The original color image is transformed first into a gray-level image. Subsequently, the gray-level image is transformed into a binary image and the threshold is determined using Otsu's method [21]. The spray boundary is detected in the binary image and the spray angle is calculated, as shown in Fig. 4. The spray boundary is searched in a rectangular frame with a height of 150 pixels, where the spray has developed well in the radial direction and the spray boundary is clear enough to detect. The width of the frame is given according to the spray scale under different operating conditions. Considering the scale of the spray image is 0.1095mm/pixel, the spray boundary is measured with a height of 10.0 mm from a distance

of 8.2 mm from the pintle tip (responding to the position where the jet is ejected from the radial orifices). For each operating condition, 1000 images are processed to obtain the time-averaged spray angle.

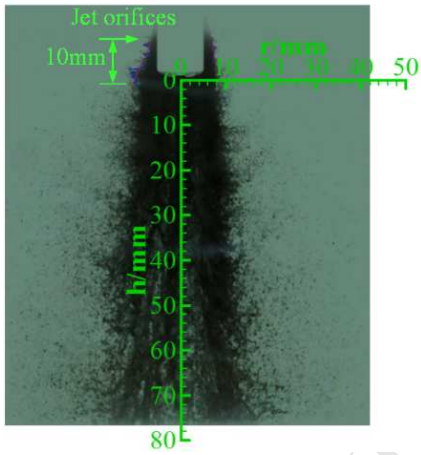


Fig. 4: Determination of the spray angle

The data obtained by the PDA system includes the diameter, axial velocity and the radial velocity of all the droplets at every single measurement point. The mean velocity and SMD are calculated for certain duration (usually the shortest measuring time of all the measurement points). A validation work is firstly done to research the influence of sample size on the statistical results. The experimental data at six measurement points is selected. For each single point, SMD and the mean axial velocity is statistically processed using all the samples, 50% samples, 20% samples and 10% samples, successively. The comparison in Fig. 5 indicates that the relative error of SMD is less than 5% and the relative error of the axial velocity is less than 2%. Therefore, the sample size hardly affects the statistical results.

ACCEPTED MANUSCRIPT

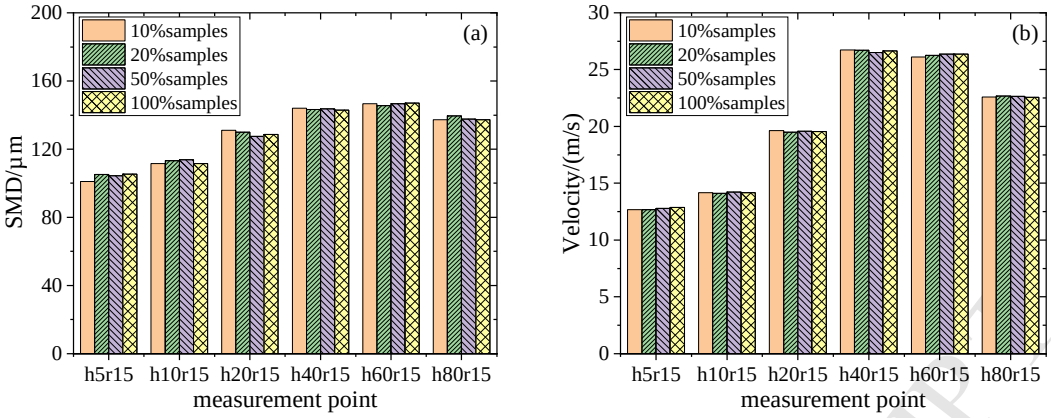


Fig. 5: Validation on the influence of sample size on the statistical results

3. Results and discussion

3.1 Spray pattern

3.1.2 Description for the spray pattern

The spray patterns under different operating conditions are displayed in Fig. 6. The spray of pintle injector is shaped like a cone with a liquid center and a rough boundary, which differs from that of a hollow cone spray generated by a swirl injector (seen in Fig. 7(a)).

The spray generated by an impinging injector is similar to that of the pintle injector because both sprays are generated by the impingement of flows from different directions. However, the spray of the pintle injector is quasi-axisymmetric whereas the spray of the impinging injector appears different when viewed from the front or the side (seen in Fig. 7 (b)). And the spray edge of the pintle injector is less distinguished than that of the impinging injector[22]. Moreover, the spray of the impinging injector develops in an open space, whereas that of the pintle injector is restricted to a semi-open space by the pintle tip. The pintle tip may also guide part of the axial film to flow along the surface of the tip and gather in the center of the spray. Furthermore, the spray pattern undergoes an expansion process with the increase in the momentum ratio and the center of the spray becomes denser.

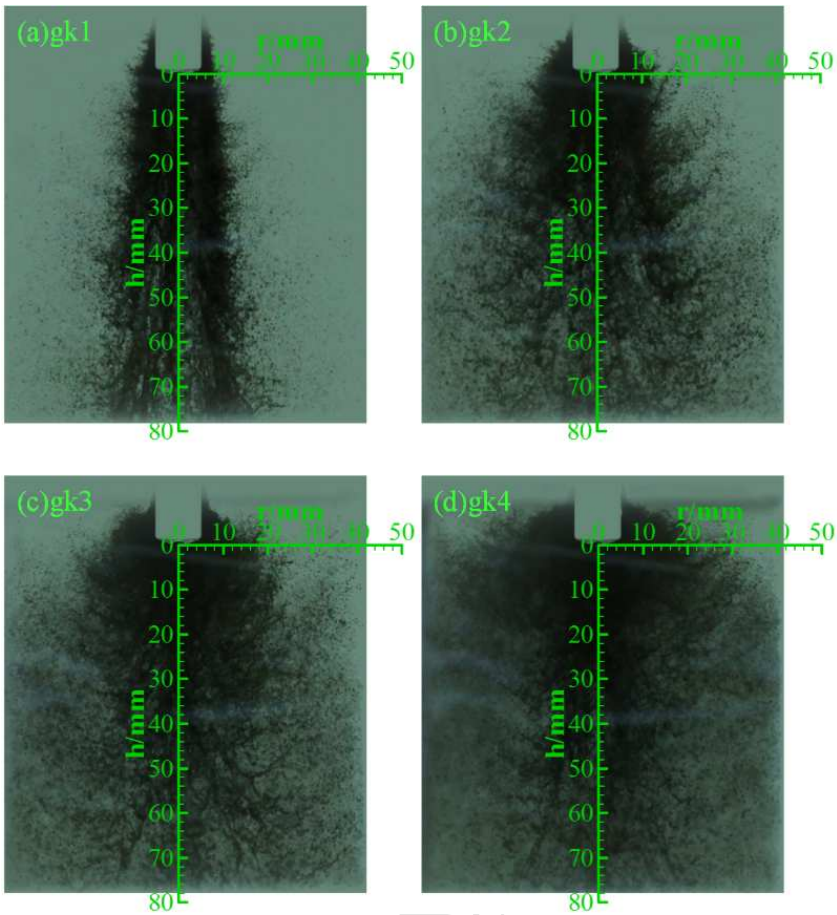
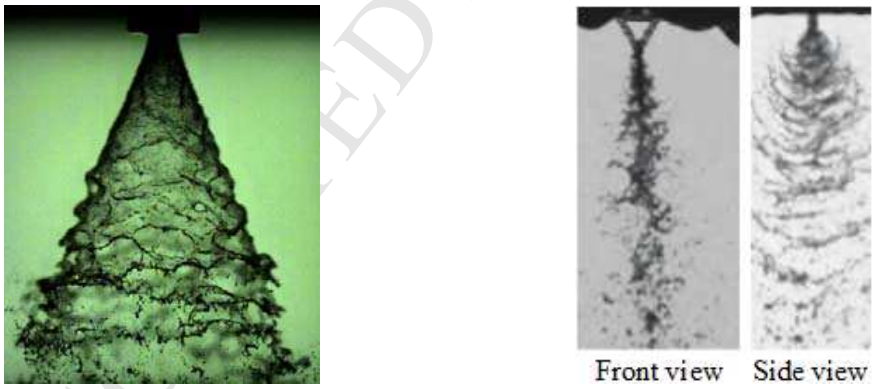


Fig. 6: Spray patterns under different operating conditions



(a) Spray pattern of a swirl injector [23] (b) Spray pattern of an impinging injector [24]

Fig. 7: The spray pattern of (a) a swirl injector and (b) a like-doublet impinging injector

The configuration of the spray is the primary concern to learn the spray characteristics of the pintle injector. Heister[1] concluded that the resultant interactions of the fluid from the radial post and the annular sleeve produced a conical spray similar to a hollow cone swirl atomizer, and Ninish[25] presented that the geometry of liquid before the formation of the spray was a hollow

conical sheet definitely at the momentum ratio of 1 and 0.7. However, the spray in this study is dense visually, thus the three-dimensional volume rate is calculated according to the PDA data, and the result is given in Fig. 8.

It is noted that the spray boundary is redefined by calculating the volume rate of every single measurement points. For example, the volume rate at r=0~10 mm and r=40 mm is relatively low compared with the central spray, as shown in Fig. 9, thus the inner and outer boundary of the spray is redefined as 15 mm and 30 mm, respectively.

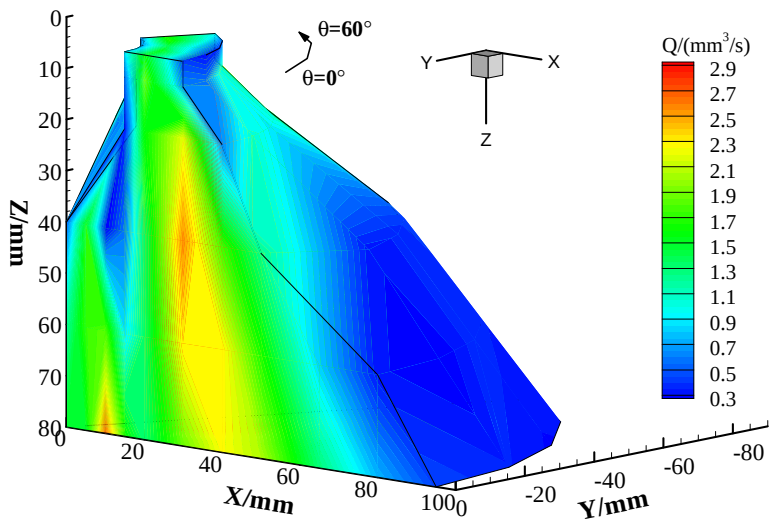


Fig. 8: The three-dimensional distribution of the volume rate (Q, hereinafter the same) of the spray at operating condition gk2 (the unit of the volume rate is mm^3/s, hereinafter the same)

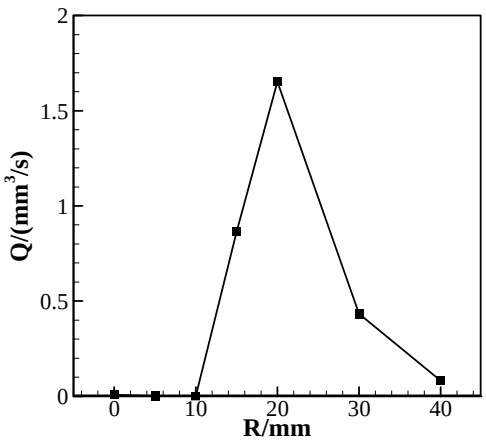
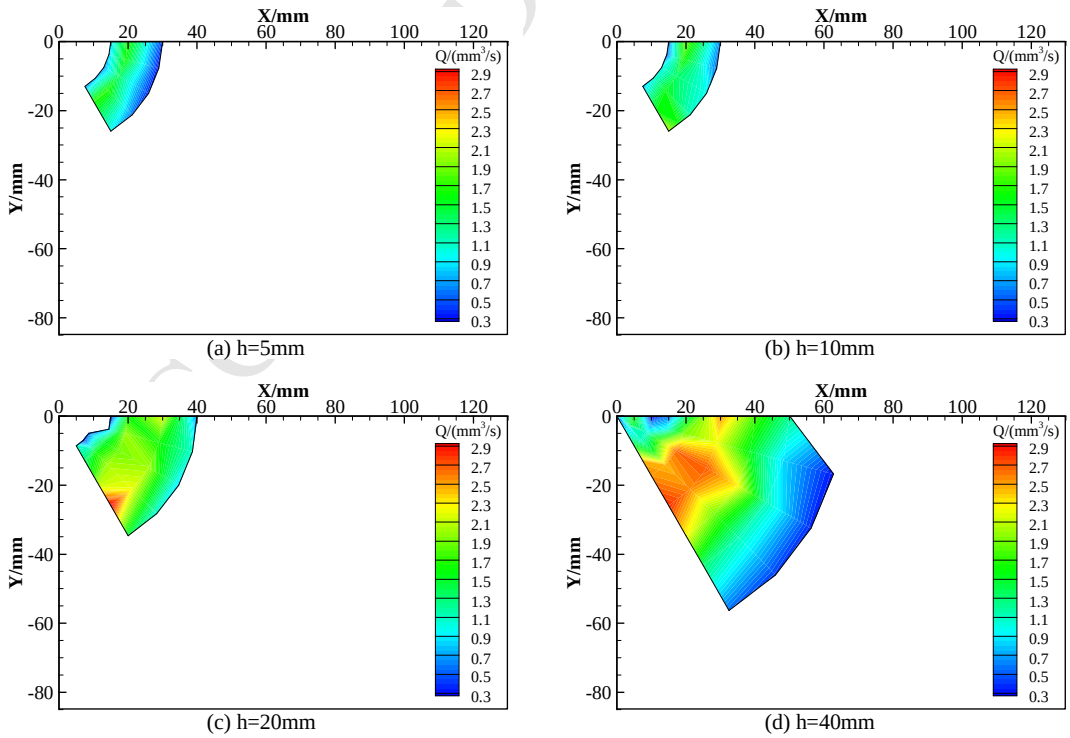


Fig. 9: The volume rate along radial direction at h=5mm, θ=0° under the operating condition gk2

The three-dimensional distribution of volume rate describes the droplet density of the whole

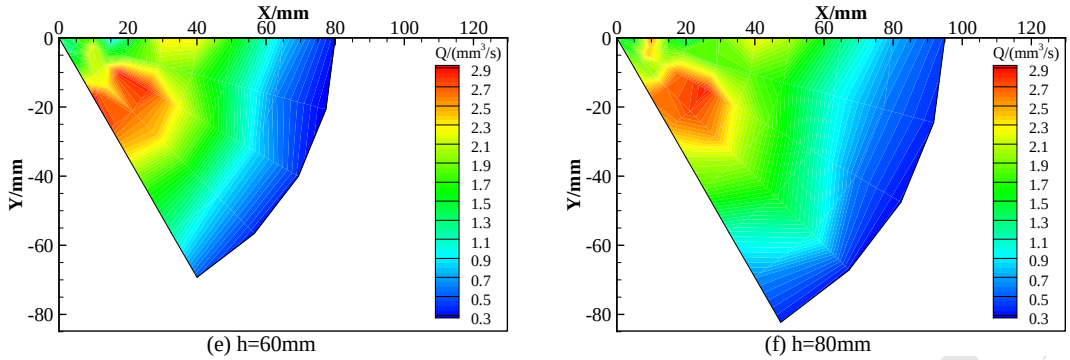
spray field and it is obvious that the volume rate is relatively low in the central and outer spray in general. For further insight, the spray development at different axial distances is plotted and shown in Fig. 10. From the near field to the far field, the hollow field shrinks and disappears; from the spray center to the spray edge, the volume rate reaches its peak and then decreases. Besides, the volume rate is unevenly distributed in the circumferential direction.

The existence and extinction of the hollow field can be attributed to the spray construction formed by the impingement of axial film and radial jet. According to the experiment results of Cheng[26] et al., the film is pushed by the jet and a “cloak” is formed. The pushing effect prevents the film from flowing down along the pintle tip, bringing about the hollow structure in the near field of the spray. However, neighboring cloaks formed by adjacent jet tends to impinge with each other and the droplets fill the hollow field, which explains why the hollow field disappears in the far field.



(a) h=5mm (b) h=10mm

(c) h=20mm (d) h=40mm



(e) h=60mm (f) h=80mm

Fig. 10: The volume rate contour at different axial distances

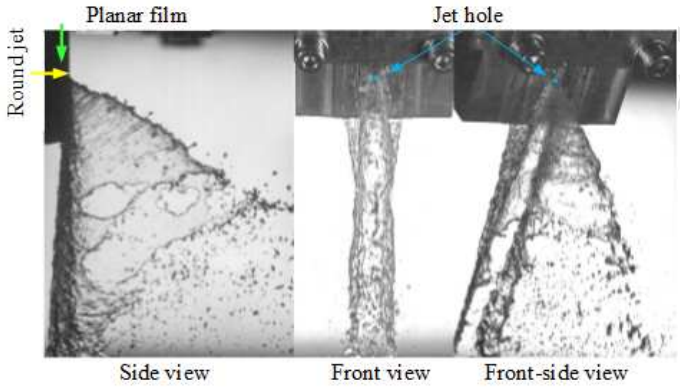


Fig. 11: Spray of the pintle injector element from different views[26]

3.1.2 Influence of the momentum ratio on the spray pattern

The influence of the momentum ratio on the spray pattern is also discussed, as shown in Fig. 12. With the local momentum ratio increasing from 0.16 to 0.93, the outer boundary of the spray field expands to a large scale and the spray angle increases. It is also noted that the outer boundary of the spray calculated by the PDA data is wider than that measured by the spray image. This is in accordance with the work of Li[27], in which he found that the penetration height of the transverse liquid jet in supersonic cross-flow obtained by PDA was larger than that of the high speed photography. The reason is that the PDA system can catch the droplets at a certain characteristic plane accurately. On the contrary, the spray image contains the overlapping effect, causing that the density of central spray is over estimated and the density of the outer spray is underestimated. Consequently, the outer boundary shrinks while the inner boundary is not distinguished in the

spray images.

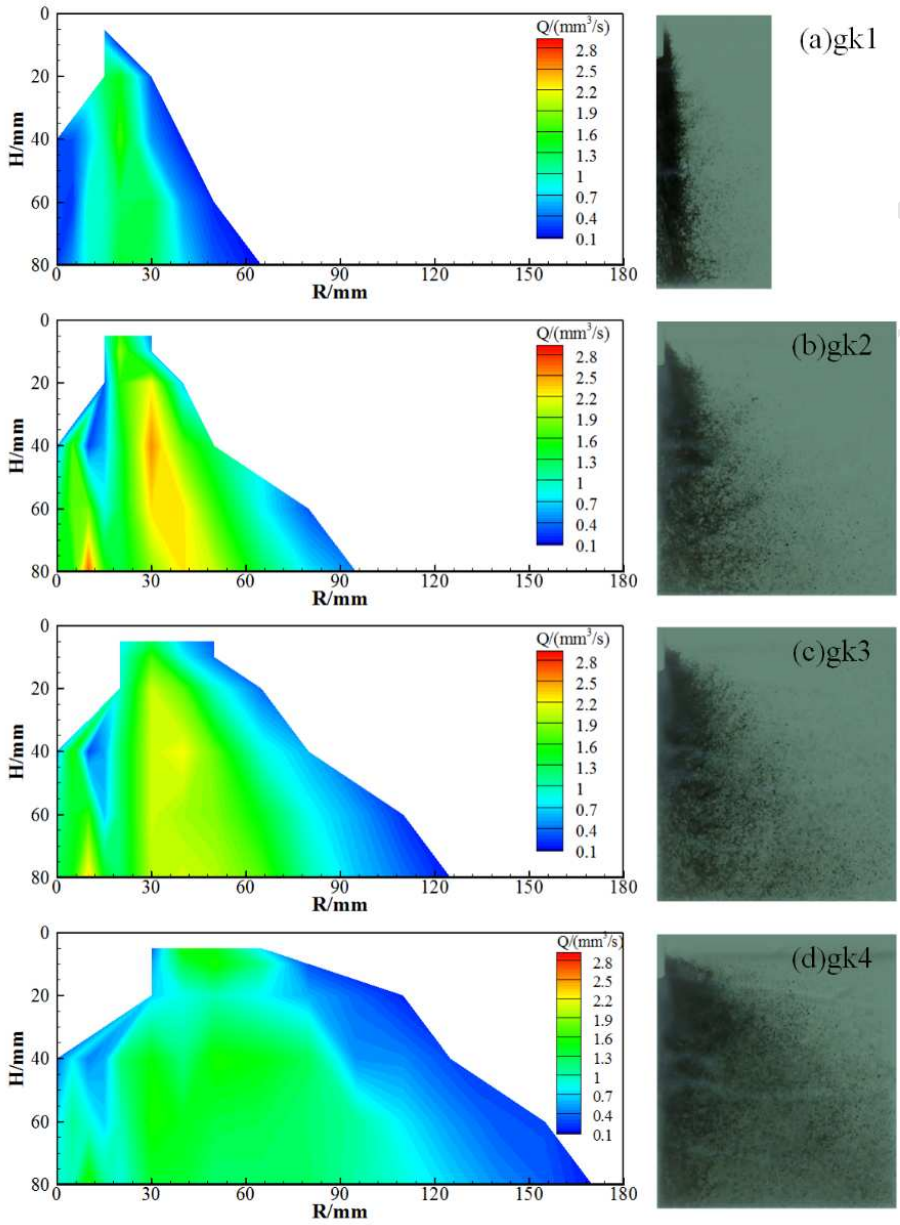


Fig. 12: The volume rate contour and corresponding spray image at different operating conditions

It is interesting that the inner boundary also expands under the higher momentum ratio, leading to the expansion of the hollow field. As mentioned before, the film is pushed into a “cloak” by the jet. To simplify the problem, the influence of the momentum ratio on the thickness expansion of the jet is neglected. Therefore, the duration before the neighboring cloaks impinge with each other is consistent. Owing to the higher radial velocity under the larger momentum ratio,

the cloak travels farther before impinging with the neighboring cloak, and the radial scale of the hollow field also expands with the increase of the momentum ratio.

3.2 Spray angle

Although the basic shape and form of the spray can be visualized by the spray image, a quantitative measurement of the spray expansion requires the calculation of the spray angle. Using the method described in section 2.4, the spray angle is measured and plotted in Fig. 13. It shows that the spray angle increases from 34° to 122° as the LMR increases from 0.16 to 0.93, which is in agreement with the expansion of the spray shown in Fig. 6.

Previous researches have indicated that the spray angle of the pintle injector is dependent on the TMR [9-13]. For example, Cheng et al. [11] proposed a prediction formula for the spray angle of liquid-liquid pintle injectors based on the simulation results and theoretical analysis. This model is based on the momentum conservation in the axial direction because the pintle wall influences the spray to some extent. The spray half angle θ' is deduced as the function of the TMR:

$$\cos\theta' = \frac{1}{1+TMR} \tag{5}$$

By replacing TMR with LMR, the prediction formula of the spray angle of pintle injector with discrete radial orifices is given in Equation(6), which agrees well with the experimental results, as shown in Fig. 13.

$$\alpha = 2a\cos\frac{1}{1+LMR} \tag{6}$$

where the LMR is defined in Equation(3).

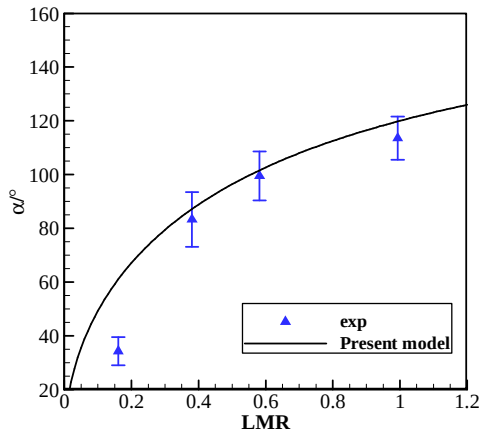


Fig. 13: Spray angle vs. local momentum ratio

Compares the spray angle obtained from different models. It is noted that TMR is replaced by LMR considering the radial channel is shaped as discrete orifices rather than continuous gap. One model mentioned by Heister [28] et al. indicates that the cone angle of the spray scales as $TMR^{0.5}$. The curve of this model in Fig. 14 is driven by equation $\alpha = 360/\pi \cdot LMR^{0.5}$. Boettcher [29] et al. studied a gas-gas pintle injector and proposed a model as follow:

$$\cos\frac{\alpha}{2} = \frac{1}{\sqrt{1+LMR^2}} \tag{7}$$

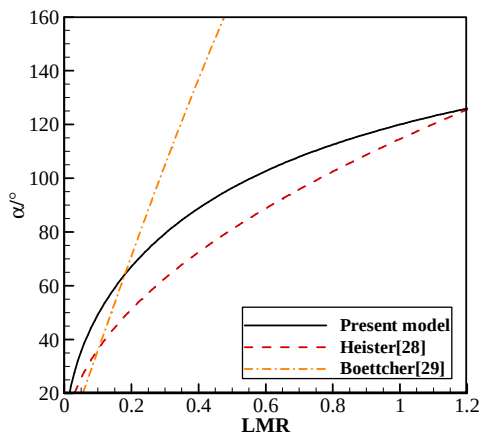


Fig. 14: Comparison of different models

As can be seen, the difference between Heister's model and the present model is less than 20° in the momentum ratio range of 0.16 to 0.93. By contrast, the spray angle predicted by Boettcher [29] et al. is relatively higher than the that of other models, which indicates that the effect of pintle wall is not negligible in the liquid-liquid pintle injector.

3.3 Diameter-velocity distribution

In order to evaluate the spray characteristics of the pintle injector, the droplet size, axial and radial velocity are measured using the PDA system.

The distribution range of the spray field is the primary concern. The diameter-velocity contour is plotted to evaluate the main scale that the spray droplets are distributed. The droplets are in the axial velocity range of 0~40 m/s and the diameter range of 0~225 μm. For all the droplets of every single measurement point per second, the number of droplets at certain axial velocity and droplet size is counted and the volume rate (Q_t) is calculated, and the results are displayed in Fig. 15. It is noted that since the number of the measurement points differs under different operating conditions, the contour is only used to describe the spray density at different diameter and radial velocity, rather than the comparison of the volume rate value under different operating conditions.

Owing that the volume rate is the cubic function of the droplet diameter, a small amount of large droplets carries a large amount of volume rate. Accordingly, the large droplets are dominant in the diameter-velocity contour. It is found that the axial velocity decreases slightly with the increase of the momentum ratio. Besides, the proportion of the large-sized droplets increases with the increase of the momentum ratio. A possible explaination is that the speed of the radial jets increases and spreads to the further region at higher momentum ratio. Owing that the spray is relatively thin and the interaction among the droplets is weak in this region, more arge droplets are generated.

The distribution of SMD under different operating conditions is analyzed in Fig. 16. Obviously, the variation trend along the radial direction changes at different axial distances. In the hollw field of h=0 mm~20 mm, SMD decreases firstly and then increases along the radial

direciton, indicating as a “V” shape. However, after the hollow field is filled, the SMD decreases first, and the variation trend changes from “*V*” shape into “*N*” shape.

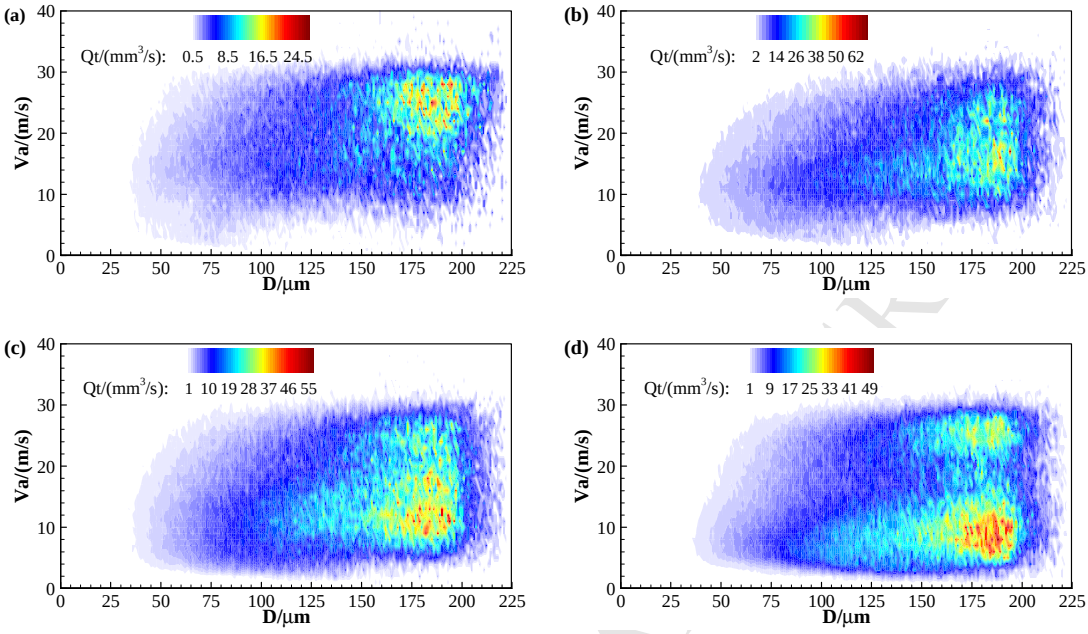


Fig. 15: Global diameter-velocity contours under (a) gk1; (b) gk2; (c) gk3; (d) gk4

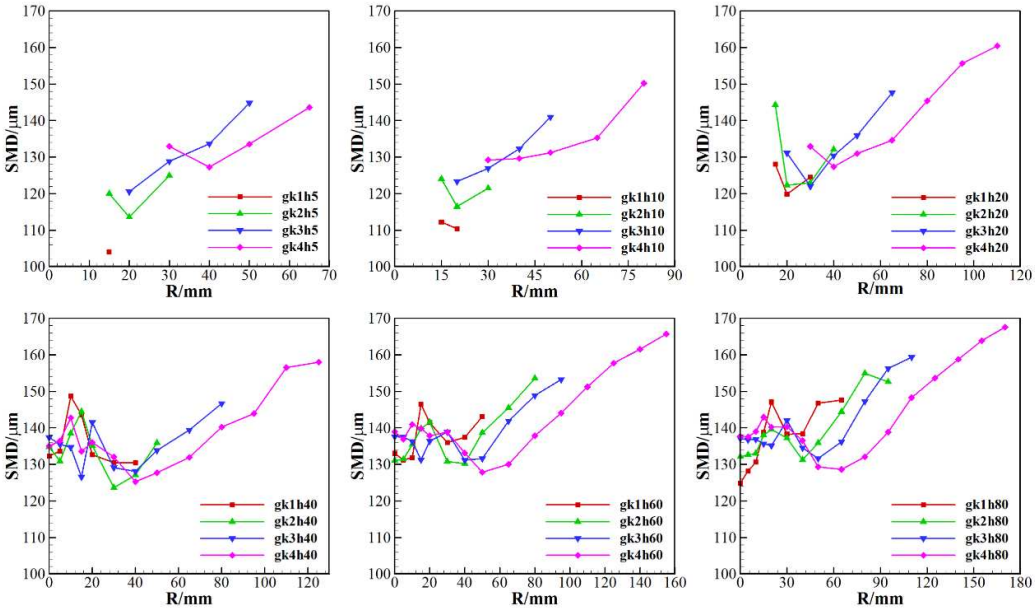


Fig. 16: Variation of SMD along radial/axial direction under the different operating conditions

The distribution of the axial and radial velocity is analyzed in Fig. 17 and Fig. 18. The variation trend of the axial velocity and radial velocity along the radial directions are similar, indicating as an inverted “*V*” shape. And that the increasing trend only exists in the solid cone

region. The radial velocity at r=0 mm is close to zero. Though the trend of the axial and radial velocity is similar, there is some difference in detail. The axial velocity only increases in a small region from r=0 mm to r=20 mm around, which responds to the bottom of the hollow region, and then decreases from r=20 mm to the spray edge. A possible explanation of the ascent stage of the axial velocity is that the filling process of the hollow field is accompanied with the gathering and collision of the flow at the spray center. This process may lead to the loss of axial velocity in the spray center. As for the radial velocity, it increases rapidly to the peak first, and then decreases slowly and slightly. It can be seen in Fig. 12 that the droplets are sparse and scattered at the outer spray. On the one hand, the droplets cannot be further atomized and SMD is rather high; on the other hand, the velocity loss is also small, thus both of the axial and radial velocity decreases mildly at the outer spray.

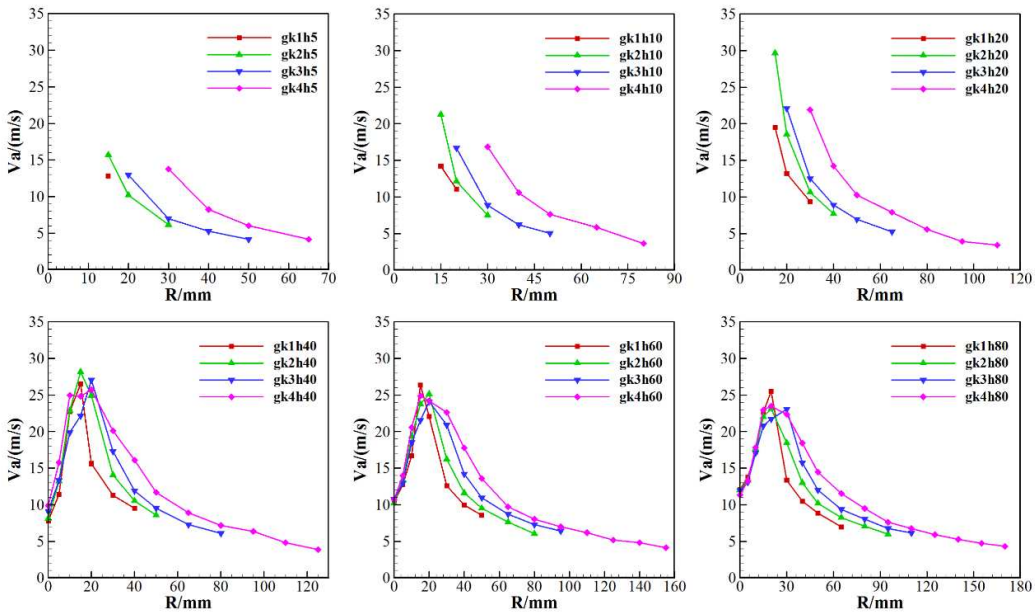


Fig. 17: Variation of axial velocity along radial/axial direction under the different operating conditions

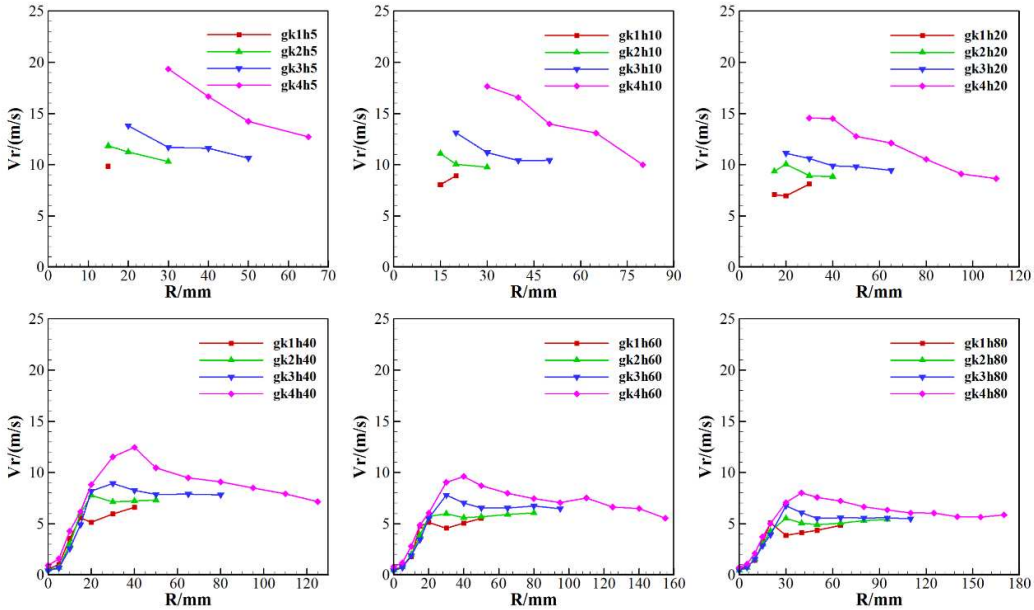


Fig. 18: Variation of radial velocity along radial/axial direction under the different operating conditions

With the increase of the momentum ratio, both of the axial velocity and the radial velocity increase. It is easy to understand that the radial velocity increases with the momentum ratio. Nevertheless, the variation trend of the axial velocity seems to be not reasonable. It is noted that the spray expands in the radial direction while travelling in the axial direction, thus the analysis and comparison of the axial velocity should be conducted at the normalized radial distance rather than the real radial distance. As shown in Fig. 19, the axial velocity decreases with the increase of the momentum ratio under the normalized radial distance in general. When the normalized radial distance is relatively small, the axial velocity under higher momentum ratio is higher. This region responds to the spray center, where the gathering and collision of the droplets may influence the velocity distribution.

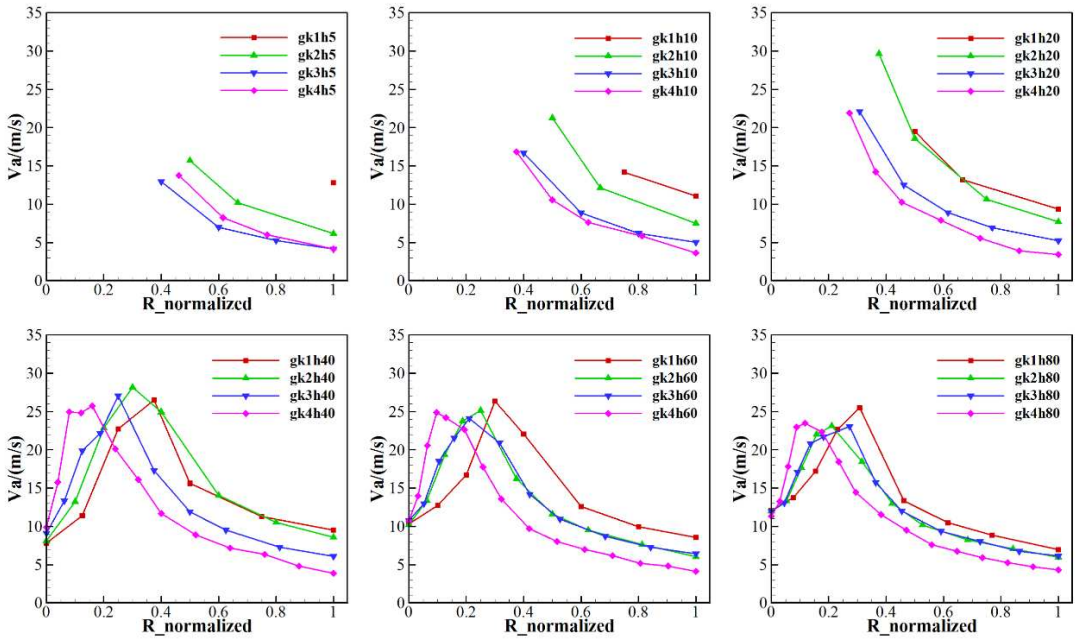


Fig. 19: Variation of the axial velocity along the normalized radial distance at the different axial distances

3.4 Spatial division of the spray

It can be seen from the previous analysis that the spray characteristics have an evident distribution feature in the radial direction. Therefore, the spray development is analyzed and characterized in the radial direction. As shown in Fig. 20, the trend of SMD, the axial and radial velocity at h=80 mm under the operating condition gk2 are analyzed as an example. According to the distribution of SMD and the axial/radial velocity, the spray field can be divided into three regions qualitatively: the spray core, the mainstream and the outer boundary.

The SMD in the mainstream is rather small due to the interaction between the film and the jet. In addition, the axial velocity decreases while the radial velocity increases along the radial direction. As for the spray core, all of the axial velocity, radial velocity and SMD increase along the radial direction, and the minimum value is at r=0 mm. This region is developed from the hollow structure in the near field. Owing to the collision of the neighboring cloaks, droplets are generated, gather and imping with each other from different directions. Considering the

impingement at r=0 mm is the strongest, the droplet size here is the least in the spray core and increases in the radial direction. The impinging effect also leads to the loss of the axial velocity. The increasing trend of the radial velocity in the radial direction attributes to the expanding of the spray. The characteristic of the outer boundary is summarized as large droplet size, low axial velocity and large radial velocity. It is thought that this region is mainly influenced by the radial jet. After the collision and interaction with the film, the droplets with radial momentum travels to the outer edge of the spray. Considering the spray density (indicated as the volume rate, seen in Fig. 8) is relatively thin in the outer boundary, the interaction among the droplets is rather weak. As a result, SMD is generally large and increases in the radial direction.

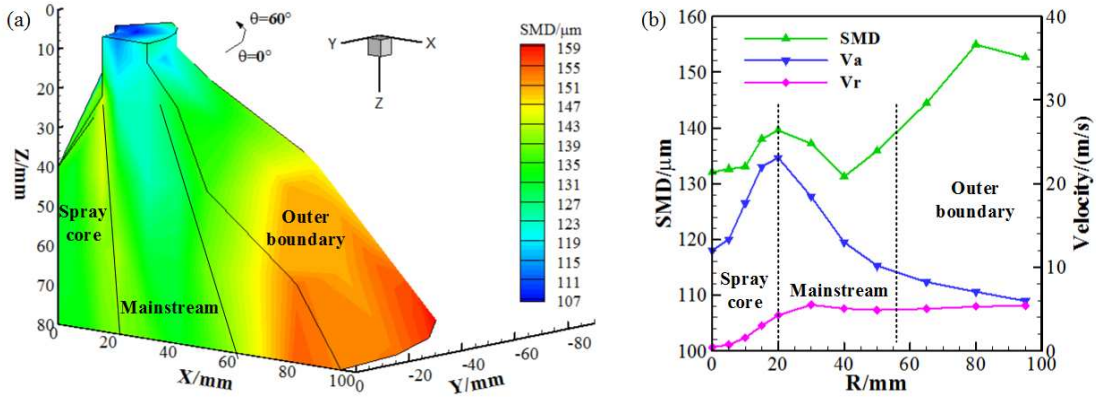


Fig. 20: Schematic diagram of (a) the three-dimensional SMD distribution; (b) the trend of SMD, the axial velocity and the radial velocity at h=80 mm under the operating condition gk2

4. Conclusion

The spray pattern, spray angle, and diameter-velocity distribution of a liquid-liquid pintle injector have been investigated experimentally. The spray distribution and spray development are analyzed. The conclusions are as follows:

a) The spray of the pintle injector is shaped as a hollow-to-solid cone with a rough boundary under the present operating conditions. A hollow structure is generated in the near field and disappears in the far field. The formation and impingement of the “cloak” structure (which is

generated by the single jet pushing the film) may attribute to the existence and extinction of the hollow field.

b) While increasing the momentum ratio by increasing the mass flow rate of the jet under constant mass flow rate of the film, the outer boundary is enlarged, leading to a higher spray angle. In detail, the spray angle ranges from 34° to 122° for the LMR in the range of 0.16 ~ 0.93. Besides, the range of the hollow field expands under the increasing momentum ratio.

c) The droplets are in an axial velocity in the range of 0~40 m/s and a diameter in the range of 0~225 μm under the present operating conditions. It is found that the droplet size increases mildly and the axial velocity decreases slightly with the increase in the momentum ratio.

d) The droplet size, axial velocity, and radial velocity exhibit an evident trend along the radial direction. The SMD curve displays as an "*N*" shape in the radial direction, whereas the axial and radial velocity curves display as inverted "*V*" shapes. It is noted that the first increasing stage is not observed in the hollow field.

e) The interaction of the radial jet flow and axial film flow dominates the spray development, and the impinging and interaction of the liquid in the spray center also has an impact. According to the distribution of SMD and the axial/radial direction, the spray of the liquid-liquid pintle injector is divided into three regions along the radial direction: the spray core, the mainstream, and the outer boundary.

Acknowledgment

The authors would like to express their sincere thanks for the support of the Chinese Program for New Century Excellent Talents in University (Grant No. NCET-13-0156), the National Natural Science Foundation of China (No. 11472303 and 11402298) and the National Basic Research

Program of China (Grant No. 613239).

References

[1] S.D. Heister, Pintle Injectors, in: N. Ashgriz (Ed.) Handbook of Atomization and Sprays, Springer Scienceþ Business Media, 2011.

[2] G.A. Dressler, J.M. Bauer, TRW Pintle Engine Heritage and Performance Characteristics, in: 36th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, American Institute of Aeronautics and Astronautics, Huntsville, Alabama, 2000.

[3] R. Gilroy, R. Sackheim, The Lunar Module Descent Engine – A Historical Perspective, in: 25th AIAA/SAE/ASME Joint Propulsion Conference, 1989.

[4] G. Elverum, P. Staudhammer, J. Miller, A. Hoffman, R. Rockow, The Descent Engine for the Lunar Module, in: AIAA 3rd Propulsion Joint Propulsion Specialist Conference, 1967.

[5] B.L. Austin, S.D. Heister, Charicterization of Pintle Engine Performance for Nontoxic Hypergolic Bipropellants, in: 38th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, Indianapolis, Indiana, 2002.

[6] B.J. Austin, Development of a Pintle-Based Engine for Nontoxic Hypergolic Propellants, in: 38th AIAA/ASME/SAE/ASEE Joint Propulsion Conference, Indianapolis, Indiana, July, 2002.

[7] B. Austin, J. Matthews, S.D. Heister, Engine/Injector Development for New Nontoxic, Storable, Hypergolic Bipropellants, in: 13th Propulsion Engineering Research Center Annual Symposium on Propulsion, Huntsville, 2001.

[8] B. Austin, S.D. Heister, W.A. Anderson, Development of Pintle and Splashplate Injectors for Nontoxic, Storable, Hypergolic Bipropellants, Journal of Propulsion and Power, 21 (2005) 627-635.

[9] M. Son, K. Yu, J. Koo, O.C. Kwon, J.S. Kim, Effects of momentum ratio and Weber number on spray half angles of liquid controlled pintle injector, Journal of Thermal Science, 24 (2015) 37-43.

[10] M. Son, K. Yu, K. Radhakrishnan, B. Shin, J. Koo, Verification on Spray Simulation of a Pintle Injector for Liquid Rocket Engine, Journal of Thermal Science, 25 (2016) 90-96.

[11] P. Cheng, Q. Li, S. Xu, Z. Kang, On the prediction of spray angle of liquid-liquid pintle injectors, Acta Astronautica, 138 (2017) 145-151.

[12] R.J. Santoro, C.L. Merkle, Main Chamber and Preburner Injector Technology, in, The Pennsylvania State University, 1999.

[13] R.D. Woodward, K.L. Miller, V.G. Bazarov, G.F. Guerin, S. Pal, R.J. Santoro, Injector Research for Shuttle OMS Upgrade Using LOX/Ethanol Propellants, in: 34th AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, 1998.

[14] M. Son, K. Radhakrishnan, J. Koo, O.C. Kwon, H.D. Kim, Design Procedure of a Movable Pintle Injector for Liquid Rocket Engines, Journal of Propulsion and Power, 33 (2017) 858-869.

[15] K. Radhakrishnan, M. Son, K. Lee, J. Koo, Effect of injection conditions on mixing performance of pintle injector for liquid rocket engines, Acta Astronautica, (2018).

[16] N.N. Smirnov, V.F. Nikitin, V.R. Dushin, Y.G. Filippov, V.A. Nerchenko, J. Khadem, Combustion onset in non-uniform dispersed mixtures, Acta Astronautica, 115 (2015) 94-101.

[17] N.N. Smirnov, V.B. Betelin, A.G. Kushnirenko, V.F. Nikitin, V.R. Dushin, V.A. Nerchenko, Ignition of fuel sprays by shock wave mathematical modeling and numerical simulation, Acta Astronautica, 87 (2013) 14-29.

[18] V.B. Betelin, N.N. Smirnov, V.F. Nikitin, V.R. Dushin, A.G. Kushnirenko, V.A. Nerchenko, Evaporation and ignition of droplets in combustion chambers modeling and simulation, Acta Astronautica, 70 (2012) 23-35.

[19] Z. Kang, Q. Li, P. Cheng, X. Zhang, Z.-g. Wang(*), Effects of self-pulsation on the spray characteristics of gas-liquid swirl coaxial injector, Acta Astronautica, 127 (2016) 249-259.

[20] W. Liyin, Breakup and Atomization Mechanisim of Liquid Jet in Supersonic Crossflows, in, National University of Defense Technology, Changsha, 2016, pp. 141.

[21] N. Otsu, A Threshold Selection Method from Gray-Level Histograms, 9 (1979) 62-66.

[22] Q.-f. Fu, R.-z. Duan, K.-d. Cui, L.-j. Yang, Spray of gelled propellants from an impinging-jet injector under different temperatures, Aerospace Science and Technology, 39 (2014) 552-558.

[23] L. Juan, Experimental and Numerical Simulation of the Breakup Process of Swirling Conical Liquid Sheet, in, National University of Defense Technology Changsha, 2012, pp. 166.

[24] L. Jia-nan, F. Jun, Y. Weidong, L. Changbo, Quasi-Direct Numerical Simulation on Atomization Characteristics of Impinging Jets Injector, Journal of Propulsion Technology, 37 (2016) 713-725.

[25] S. Ninish, A. Vaidyanathan, K. Nandakumar, Spray characteristics of liquid-liquid Pintle injector, Experimental Thermal and Fluid Science, 97 (2018) 324-340.

[26] P. Cheng, The Dynamics of Spray Combustion in Variable Thrust Rocket Engines, in, National University of Defense Technology, Changsha, China, 2018.

[27] L. Chun, Structure characteristic of transverse liquid jet in supersonic crossflow, in, Nation University of Defense Technoloty, Changsha, 2013.

[28] S. HEISTER, T. NGUYEN, A. KARAGOZIAN, Modeling of liquid jets injected transversely into a supersonic crossflow, in: 26th Aerospace Sciences Meeting, 1988.

[29] P. Boettcher, L.G. Mikellides, D. Vaughan, J.E. Shephard, J. Damazo, Visualization of tranverse annular jets, in: 62nd Annual Meeting of the APS Division of Fluid Dynamics, Minneapolis, Minnesota, 2009.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Highlights:

- The spray displays as a hollow-to-solid cone with a rough boundary
- The outer boundary and the range of hollow structure expand with momentum ratio
- The spray characteristics exhibit an evident trend in the radial direction
- The spray is divided into three regions from the spray center to the edge

ACCEPTED MANUSCRIPT